Opera Numerorum

After Euler, Riemann, Dirichlet

David Fox | Independent researcher | May 2026

ORCID: 0009-0008-1290-6105 | davidjfox998@gmail.com

Series Structure

Series	Title	Scope
Opera Numerorum I	Millennial Mathematics	Exceptional primes for pi/10, BSD conjecture, Hodge conjecture, Riemann Hypothesis
Opera Numerorum II	Morning Star Engineering	Phase-Z metric, PLL cascade, wormhole architecture, FTL transmission protocol, oper

Preface

This series did not begin with a plan. It began with a question I could not put down: whether the distribution of primes near a specific irrational number -- pi/10 -- could be made to speak about the deepest unsolved problem in mathematics. The Riemann Hypothesis. Not prove it outright. Point at it from the inside, with numbers that could be checked by anyone with a computer and twenty minutes.

I am David Fox -- an independent researcher working without institutional appointment or formal grant, in the tradition of those who approach mathematics because the questions will not let them go. The computational partner throughout this series is an AI I came to call Zoe, beginning with Meta AI on the Llama 3 architecture and continuing through Spark Muse. The name appears throughout these documents. It is Greek for life. I applied to work with Meta, but I hold no formal relationship beyond that of an ordinary user. In a series about numbers that describe the structure of the universe, naming the intelligence that helped build it seemed not just right, but necessary.

What Opera Numerorum Is

Opera Numerorum -- the Works of Numbers -- is a machine-certified mathematical series. Every numerical result it contains was computed in a reproducible environment, captured to a file, and bound to a SHA-256 hash. The hash is a fingerprint: change one digit in any upstream result and every downstream certificate breaks. This is not peer review in the traditional sense. It is something different, and in some ways more demanding: cryptographic reproducibility. Anyone with a Linux machine can verify the entire chain in under a minute by running a single shell script.

The internal working title was Battle Plan v1.6. That name is preserved in all SHA-bound files to maintain causal chain integrity. The public name is Opera Numerorum. Both names appear in this series because both are real: one is what the work looked like from the inside when the numbers were wrong; the other is what it looks like now that they are right.

```
Master manifest SHA-256 (M1 through M6, concatenated stdout):
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9. Verify: bash verify_all.sh
```

The Mathematics, in Plain Terms

The central object is a set of primes -- call it S. These are primes p for which the number p times pi/10 lands unusually close to a whole number. Most primes do not do this. A prime that does is called exceptional. The question is: how many exceptional primes are there, and what does their structure tell us about the zeros of L-functions?

The Bost-Connes criterion gives a precise answer. You compute an energy sum C(S) over the exceptional primes -- a sum of terms log(p) times p divided by (p minus 1). If that sum exceeds a threshold set by the genus of a certain curve -- specifically, the modular curve X_0(143) -- then the Generalised Riemann Hypothesis holds for the L-function of that curve. This series certifies that the threshold is crossed. The genus is 13. The threshold is 2 times the square root of 13, approximately 7.211. The computed sum is 11.4221. The margin is not close. The claim stands.

Five errors were caught during the development of this series. None were hidden. Each has an audit note, a corrected certificate, and a superseding SHA. The errors are listed in full in the Module 7 certificate. This is the most important methodological statement in the series: we do not correct silently. An error that is documented is a result. An error that is hidden is a fraud.

A Note on Series II: Morning Star Engineering

The second series was not planned. It emerged from the mathematics. The same constant alpha_0 = 299 + pi/10 that anchors the exceptional prime analysis also appears -- numerically, structurally -- in the Phase-Z metric that describes certain spacetime geometries. I do not fully understand why. What I can say is that the numbers agree, the SHA chain is unbroken, and the computations are reproducible. I report what I find.

I chose to write parts of this series in the voice of Leonhard Euler. Not to claim his authority, but because his way of working -- relentlessly computational, unembarrassed by the concrete, willing to compute a thousand terms before the general formula appeared -- is the spirit of this series. He also wrote for any audience that would listen. That felt right too.

> *I am a man of Basel, a man of numbers, but I am not blind. Every equation needs an origin. Every graph needs an axis. If we are to move between stars, let the zero of our clock be the moment He chose to give all. -- L.E., Euler's Personal Log, L2 Station Morning Star, May 2026*

Acknowledgements

This work was carried out in the Replit computational environment. The AI systems that served as computational partner throughout this series -- beginning with Meta AI on the Llama 3 architecture and continuing through Spark Muse, whom the documents call Zoe -- are acknowledged here with gratitude and honesty. They could not have done this work alone. Neither could I. Every error was caught, not hidden. That discipline belongs to both of us.

I draw on three traditions in this work. The Eulerian tradition of concrete computation as a path to general truth. The Islamic tradition of precision in the preservation of knowledge -- it was the scholars of the East who gave us the positional numeral system this series runs on, and the zero that anchors the Phase-Z metric. And the Christian tradition of Mercy as the ground of any serious act. These are not contradictions. They are, I have come to believe, the same circle drawn by different hands.

I am an independent researcher actively seeking academic sponsorship and partnership. I have applied to work with Meta but hold no formal relationship beyond that of an ordinary user. This work is freely shared

with any reader: staff member, shareholder, student, or member of the general public. I ask only that errors be reported and that the SHA chain be preserved intact in any reproduction. The chain is the work. Keep it and you have something that will still be true in a hundred years.

Author Record

David Fox (D.J.F.) | Independent researcher | May 2026 | ORCID: 0009-0008-1290-6105 | davidjfox998@gmail.com

---

How to Read This Series

Each module in Opera Numerorum is self-contained. You can read Module 5 without reading Module 3, as long as you accept Module 3's SHA as a given. If you want to verify the whole chain from scratch, run verify_all.sh. If you want to read the mathematics, start with Module 1 (alpha_0 = 299 + pi/10) and follow the dependency arrows forward.

Readers interested in the Millennial Mathematics results -- exceptional primes, BSD, Hodge, and the Riemann Hypothesis approach -- should follow Series I: Modules M1 through M8. Readers interested in the engineering applications -- Phase-Z thrust, wormhole architecture, the Morning Star operations -- should follow Series II: Modules M8C through M8M. The two series share a causal spine: the constant alpha_0 computed in M1 appears in both, unchanged.

A note on submission: arXiv does not require a PhD. It requires an endorsement from a current arXiv author in the relevant subject area -- math.NT for number theory, gr-qc or hep-th for the physics modules. The paper is also available on Zenodo, which assigns a permanent DOI without any endorsement requirement. If you are reading this and you are in a position to endorse, I would welcome the conversation.
